MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC GATE, NAND, DIGITAL, KEY CODE GDC

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389 except as modified in 3.4.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.05 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389, except the operating temperature range shall be 0° to 80°C.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GDC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/8-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

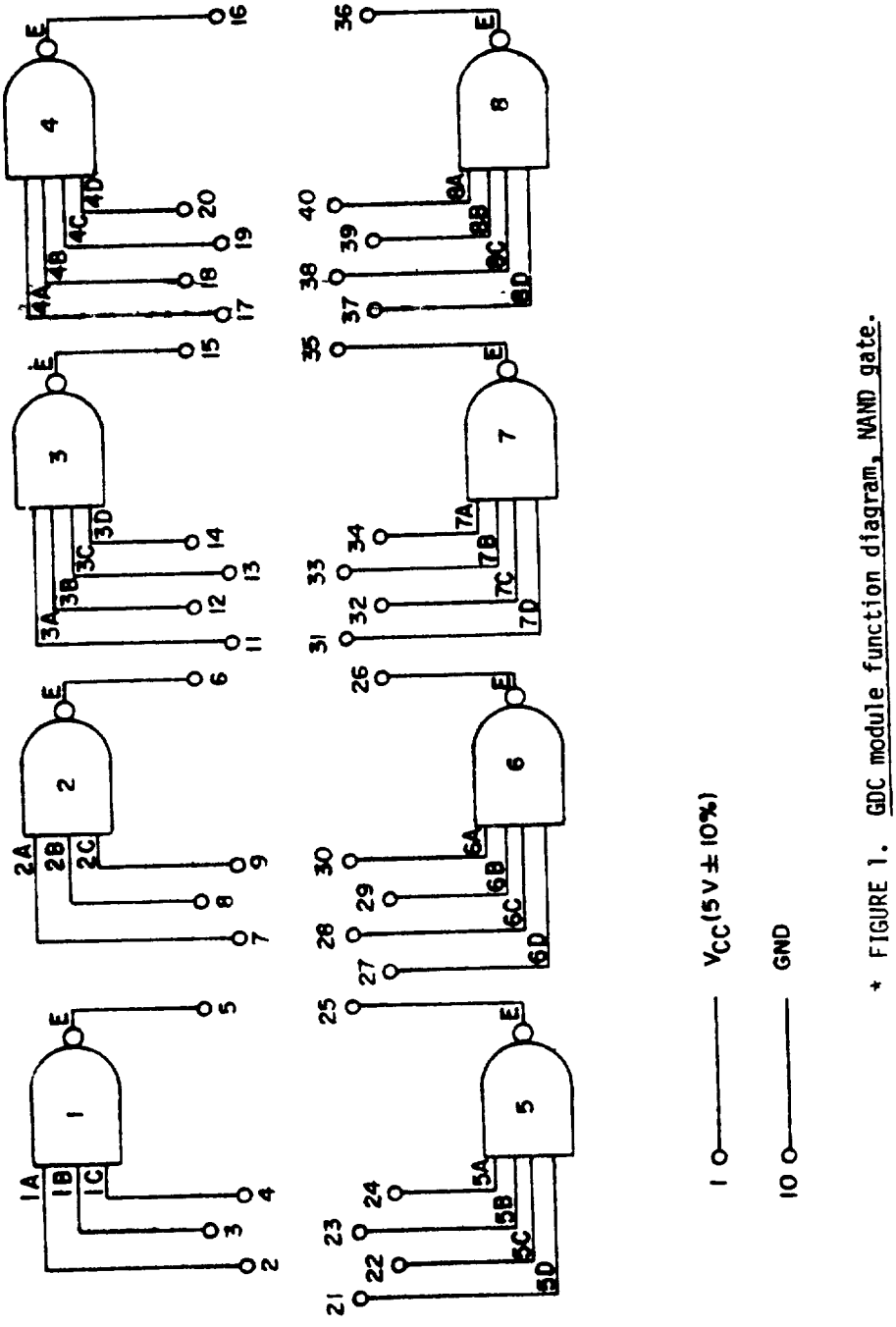
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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* FIGURE 1. GDC module function diagram, NAND gate.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 80°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A8	V_{OUT} (low loaded)	III	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
B1-B30	Input breakdown current	IV	S	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	mA
C1-C30	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D8	Output voltage, low level	VI	X	0	450	0	450	0	450	0	450	mV
E1-E30	Output voltage, high level	VI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F30	Input current, low level	VII	X	1.0	2.3	1.0	2.3	1.0	2.3	1.0	2.3	mA
G1-G84	Input current, high level	VII	X	0	120	0	120	0	120	0	120	μA
H1-H8	Output short circuit current	VII	X	50	165	50	165	50	165	50	165	mA
J1	Power supply current, low level	VIII	S	60	124	60	124	60	124	60	124	mA
J2	Power supply current, high level	VIII	S	10	40	10	40	10	40	10	40	mA
K1-K8	Transition time (t_{THL})	IX	S	3	45	3	45	3	50	3	50	ns
L1-L8	Transition time (t_{TLH})	X	S	3	45	3	45	3	60	3	60	ns
M1-M8	Propagation delay time (t_{PLH})	XI	S	3	40	3	40	3	45	3	45	ns
N1-N8	Propagation delay time (t_{PHL})	XII	S	3	50	3	50	3	55	3	55	ns
P	Power supply transient amplitude	XIII	Q	0	500	1	500	0	500	0	500	mV
Q	Isolation	XIV	S	10	∞	1	∞	10	∞	1	∞	MΩ
R1-R8	Output crosstalk, low level	XV	Q	0	800	0	800	0	800	0	800	mV
R9-R32	Output crosstalk, low level	XV	Q	0	400	0	400	0	400	0	400	mV

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 80°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S1-S8	Output crosstalk, high level	XV	Q	0	800	0	800	0	800	0	800	mV
S9-S32	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	mV
T1-T30	Input capacitance	XVI	Q	2	15	2	15	2	15	2	15	pF
U	Power supply filter capacitance	XVII	S	0.007	3.0	0.007	3.0	0.007	3.0	0.007	3.0	µF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A, B, or Q are part of a sequence of tests, the order of testing shall be as follows: Q, A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through P, R, S, and U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition U, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through P, R, S and U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through U shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/8 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658755 Part number M28787/8-1 supersedes part numbers 2658755 and M28787/8.

* 6 4 Module description. The following electrical parameters reflect the 0° to 80°C end-of-life limits from table I.

Supply voltage	5 V ±10%
Input threshold voltage	
Low level	0.8 V maximum
High level	2.0 V minimum
Input current	
Low level (V_{IL} = 0 V)	-2.3 mA maximum
High level (V_{IH} = 2.4 V)	120 µA maximum

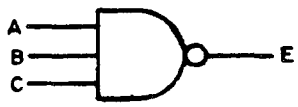
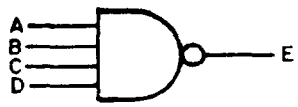
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Output voltage	
Low level	0.45 V maximum
High level	2.4 V minimum, V_{CC} maximum
Output Drive current	
Low level (V_{OL} = 0.0 V)	44 mA maximum
High level (V_{OH} = 2.4 V)	-3.0 mA maximum
Power dissipation maximum	
All outputs at low level	682 mW (outputs not loaded)
All outputs at high level	220 mW (outputs not loaded)

* 6.5 Module function. This module contains 6 four-input and 2 three-input power NAND gates.

6.5.1 Input to output relationship. Each gate shall have the input to output relatonship of table II for positive logic.

TABLE II. Input to output relationship.

Input	Output
At least one input in low state	High state
All inputs in high state	Low state
	$E = \overline{ABC}$
	$E = \overline{ABCD}$

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.5.3 Crosstalk. The module has the following crosstalk relationships.

6.5.3.1 Module crosstalk. When connected in the worst case mode of operation (all outputs but one switching simultaneously), the GDC module could exhibit substantial crosstalk due to the large switching currents of the power NAND gates. When the quiet line is either high or low, there may be as much as 800 mV of crosstalk.

6.5.3.2 Circuit crosstalk. The crosstalk on any output pin is nearly directly proportional to the number of switching outputs. For every switching output in group A (circuits 1 and 5), the noise induced on a low-or-high level quiet output in group A can be as much as 200 mV. For every switching output outside group A (circuits 2-4 and 6-8), the noise induced on a low-or-high level quiet output in group A can be as much as 100 mV. The preceding relationships also apply to group B (circuits 2 and 6), group C (circuits 3 and 7), and group D (circuits 4 and 8).

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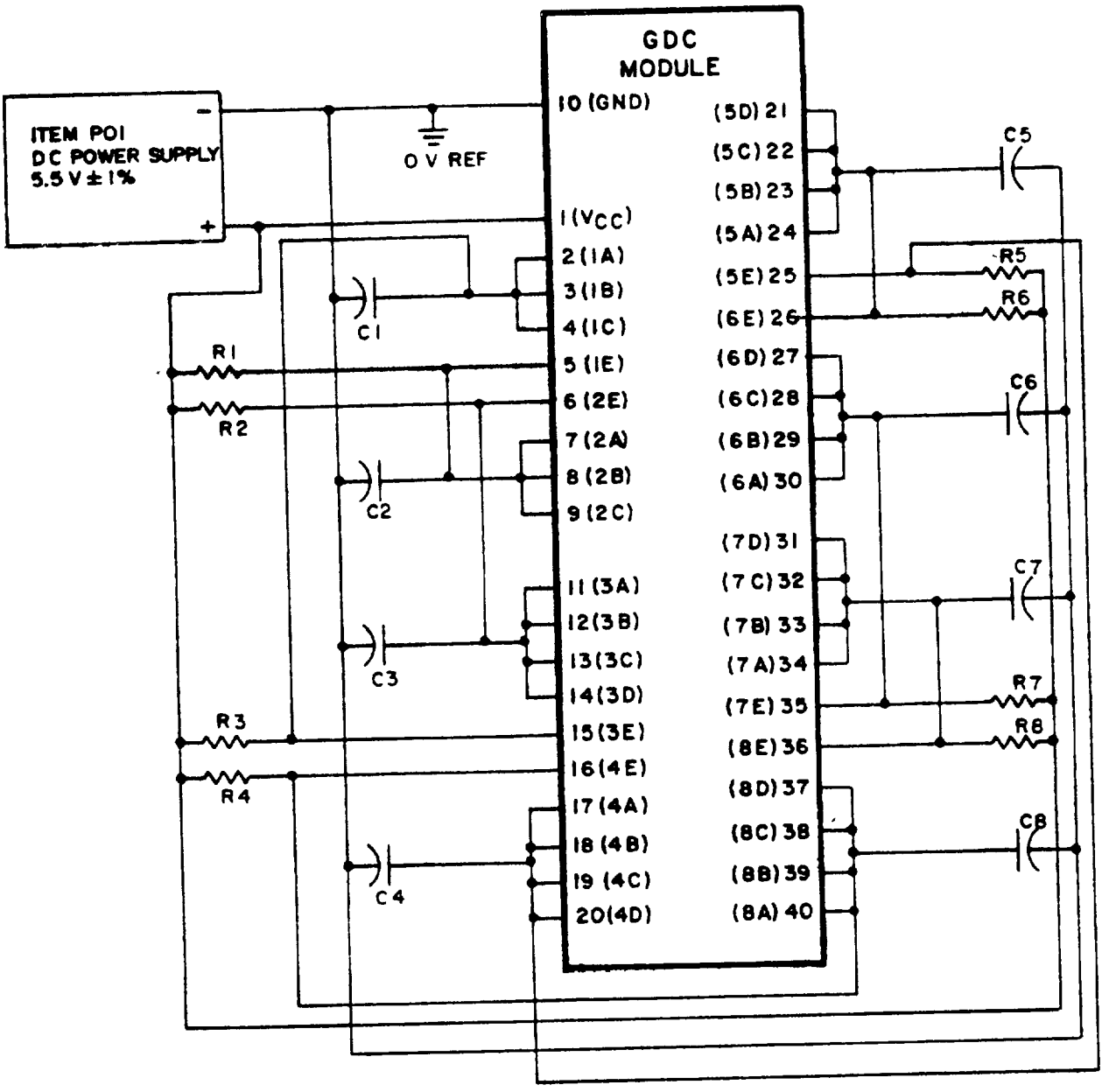
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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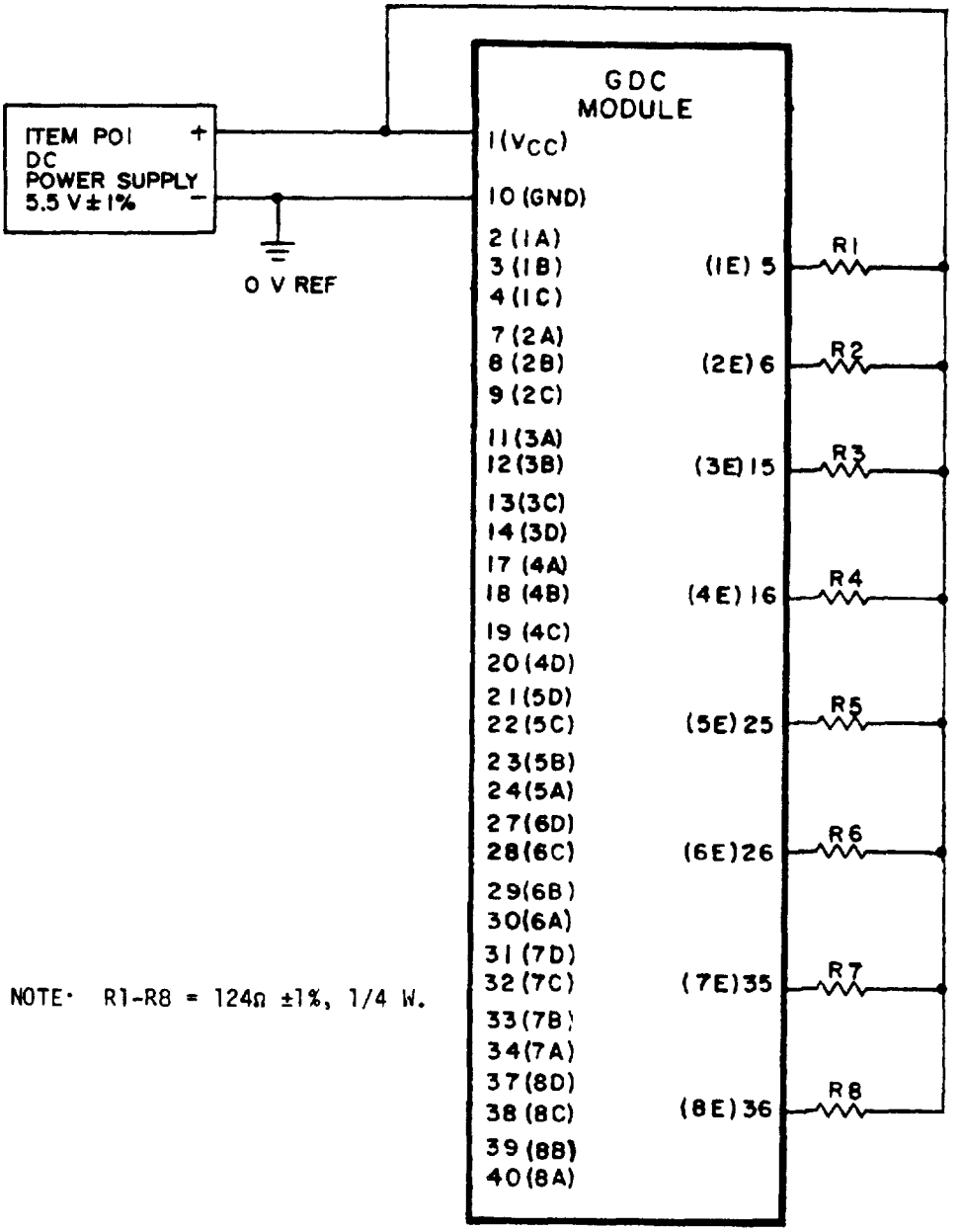
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NOTES:
1. R1-R8 = 124Ω ±1%, 1/4 W.
2. C1-C8 = 1,000 pF ±20%.

* FIGURE 2. Operating temperature and life test circuit.

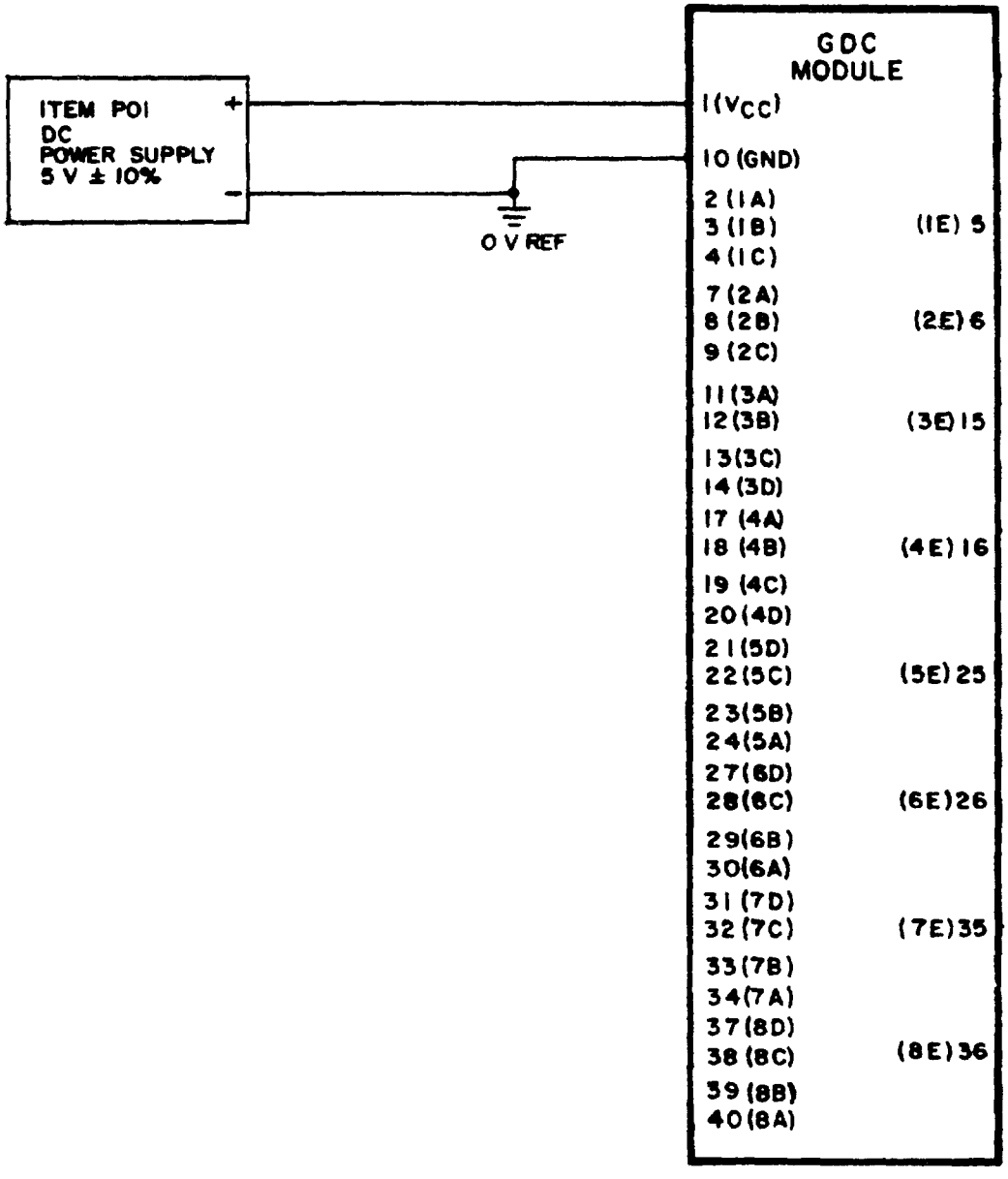
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NOTE: R1-R8 = 124Ω ±1%, 1/4 W.

* FIGURE 3. Component temperature test circuit.

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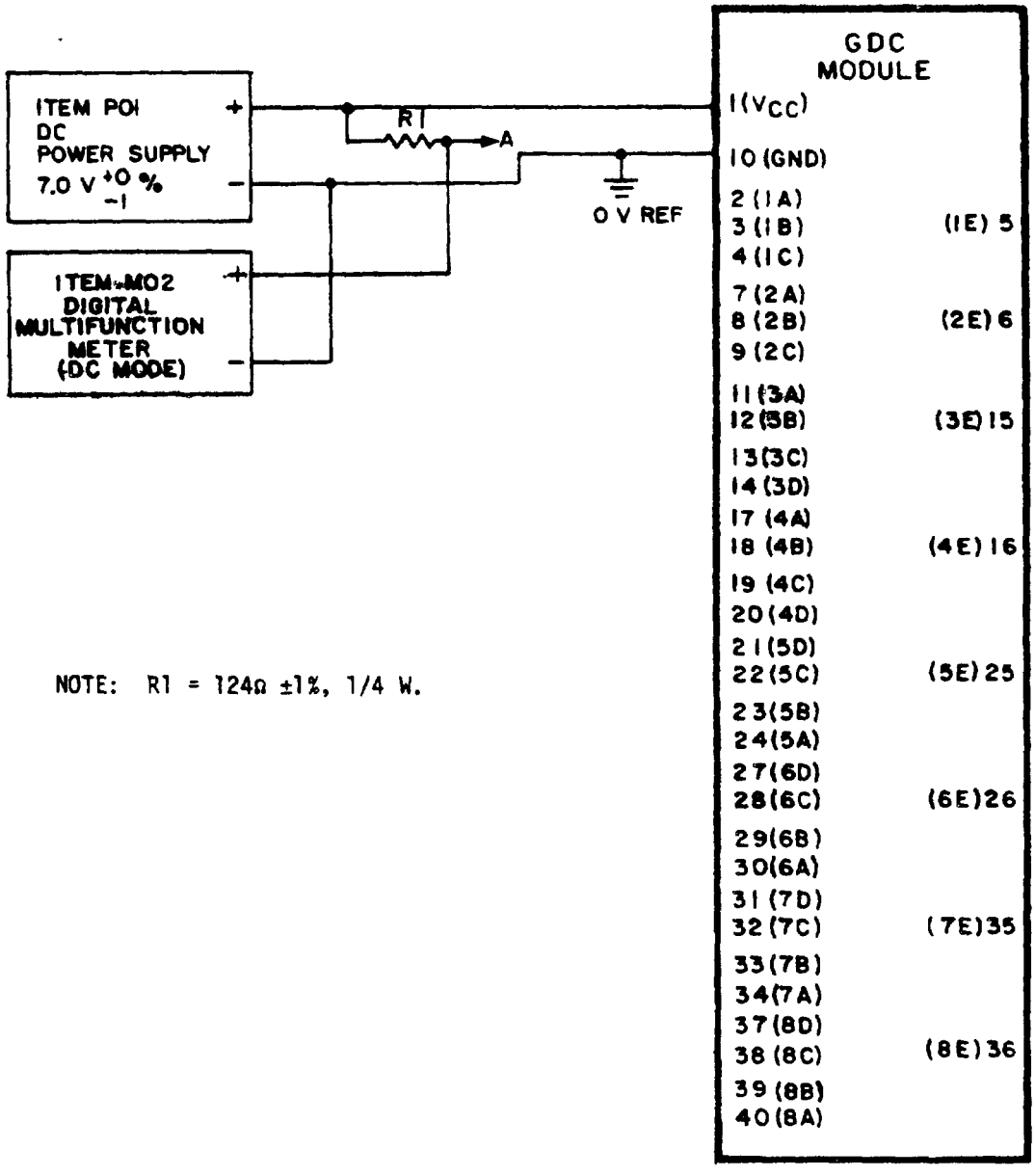
* FIGURE 4. Vibration test circuit.

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* TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	1E V_{OUT} (low loaded)	5	0.01	0.8	V	5
A2	2E	6	↓	↓	↓	↓
A3	3E	15				
A4	4E	16				
A5	5E	25				
A6	6E	26				
A7	7E	35				
A8	8E	36				

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NOTE: R1 = 124Ω ±1%, 1/4 W.

* FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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* TABLE IV. Input breakdown current test.

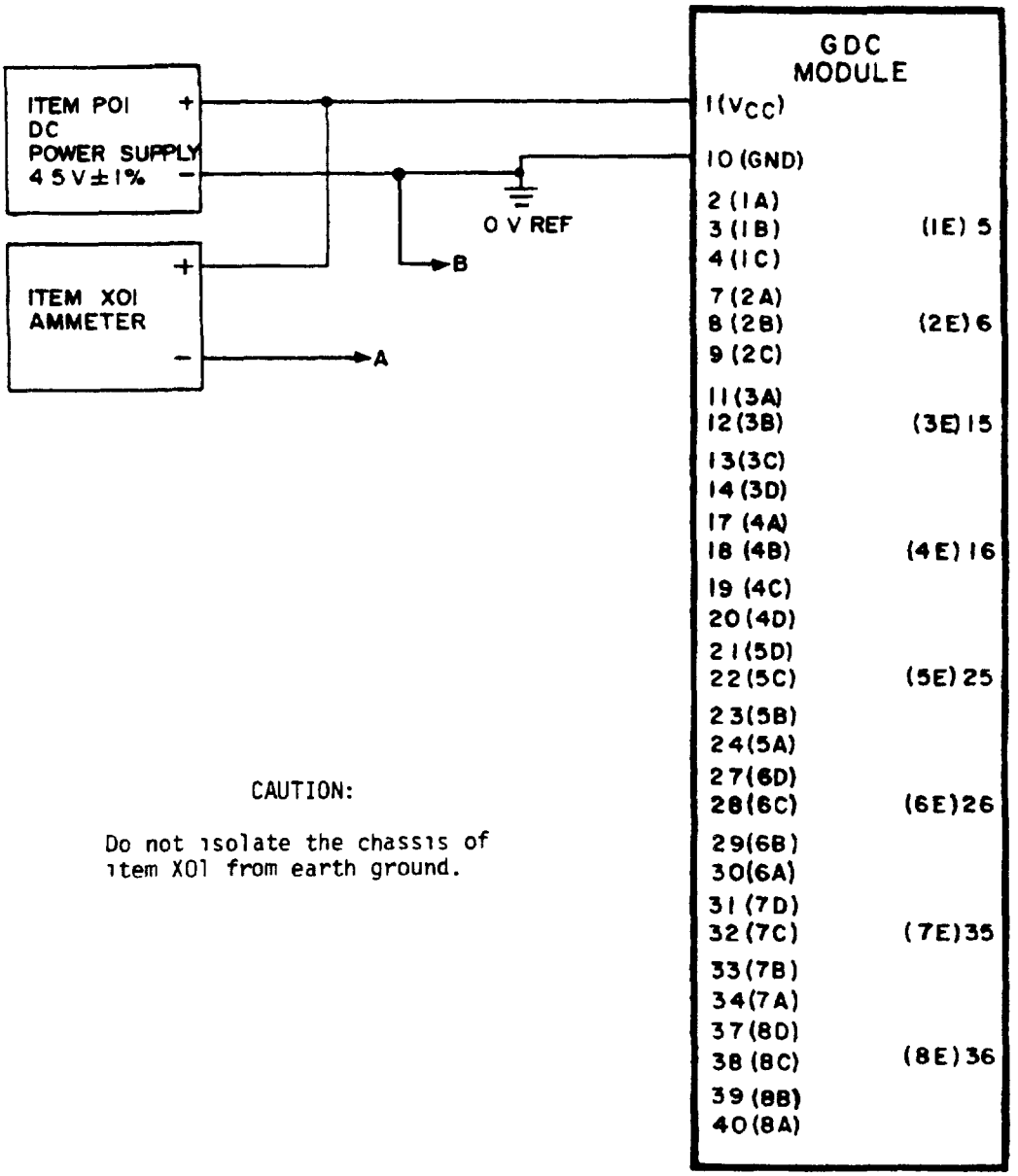
Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	1A	Input breakdown current	2	3,4	0 0	0.5	mA	6
B2	1B		3	2,4				
B3	1C		4	2,3				
B4	2A		7	8,9				
B5	2B		8	7,9				
B6	2C		9	7,8				
B7	3A		11	12,13,14				
B8	3B		12	11,13,14				
B9	3C		13	11,12,14				
B10	3D		14	11,12,13				
B11	4A		17	18,19,20				
B12	4B		18	17,19,20				
B13	4C		19	17,18,20				
B14	4D		20	17,18,19				
B15	5D		21	22,23,24				
B16	5C		22	21,23,24				
B17	5B		23	21,22,24				

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* TABLE IV. Input breakdown current test - Continued.

Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B18	5A	Input breakdown current	24	21,22,23	0.0	0.5	mA	6
B19	6D	↓	27	28,29,30	↓	↓	↓	↓
B20	6C		28	27,29,30				
B21	6B		29	27,28,30				
B22	6A		30	27,28,29				
B23	7D		31	32,33,34				
B24	7C		32	31,33,34				
B25	7B		33	31,32,34				
B26	7A		34	31,32,33				
B27	8D		37	38,39,40				
B28	8C		38	37,39,40				
B29	8B		39	37,38,40				
B30	8A		40	37,38,39				

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

* FIGURE 6. Input breakdown current test circuit for table IV.

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* TABLE V. Input clamp diode voltage test.

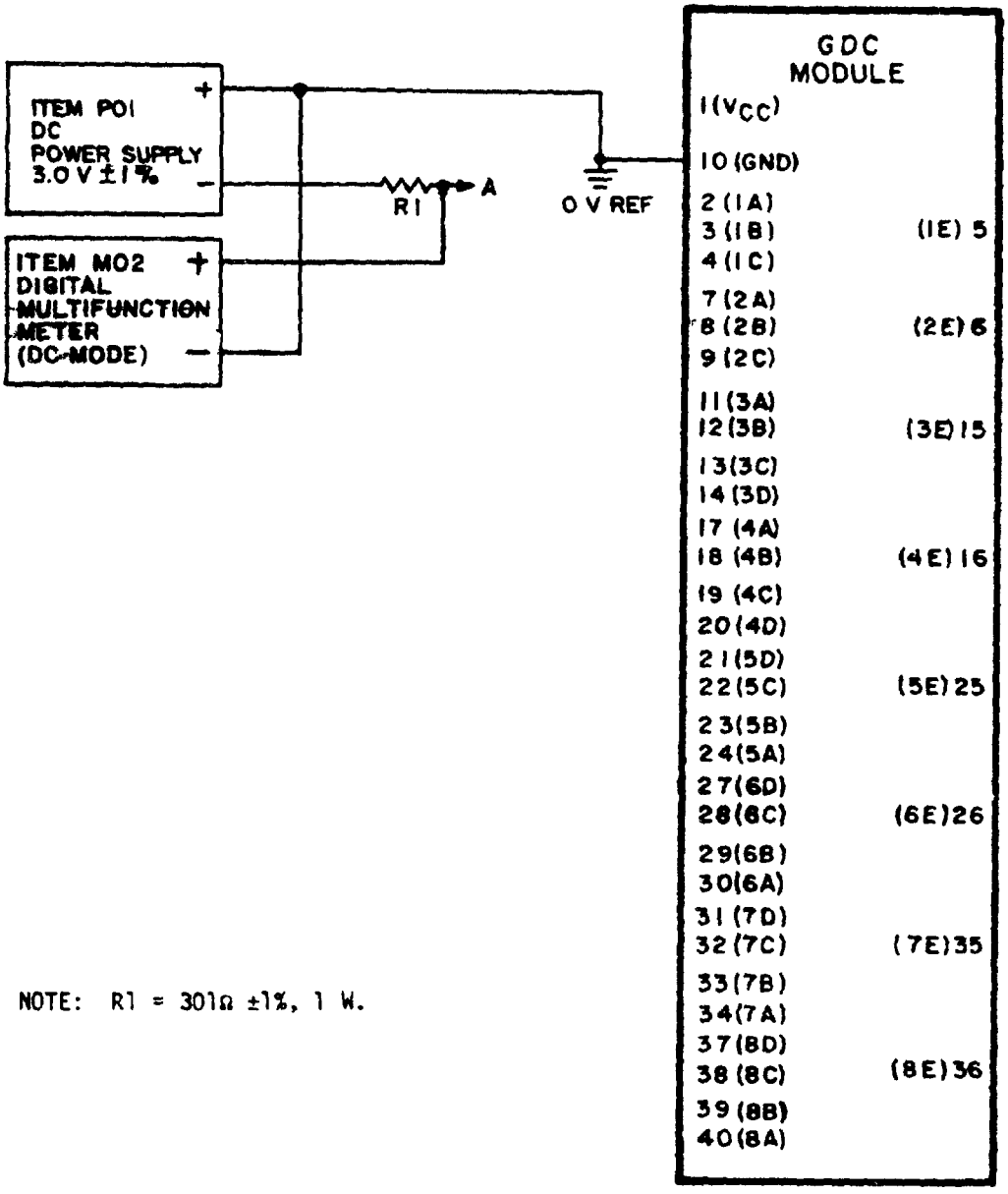
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	1A Input clamp diode voltage	2	-0.4	-1.5	V	7
C2	1B	3				
C3	1C	4				
C4	2A	7				
C5	2B	8				
C6	2C	9				
C7	3A	11				
C8	3B	12				
C9	3C	13				
C10	3D	14				
C11	4A	17				
C12	4B	18				
C13	4C	19				
C14	4D	20				
C15	5D	21				
C16	5C	22				
C17	5B	23				

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* TABLE V. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C18	5A Input clamp diode voltage	24	-0.4	-1.5	V	7
C19	6D	27				
C20	6C	28				
C21	6B	29				
C22	6A	30				
C23	7D	31				
C24	7C	32				
C25	7B	33				
C26	7A	34				
C27	8D	37				
C28	8C	38				
C29	8B	39				
C30	8A	40				

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NOTE: R1 = 301Ω ±1%, 1 W.

* FIGURE 7. Input clamp diode voltage test circuit for table V.

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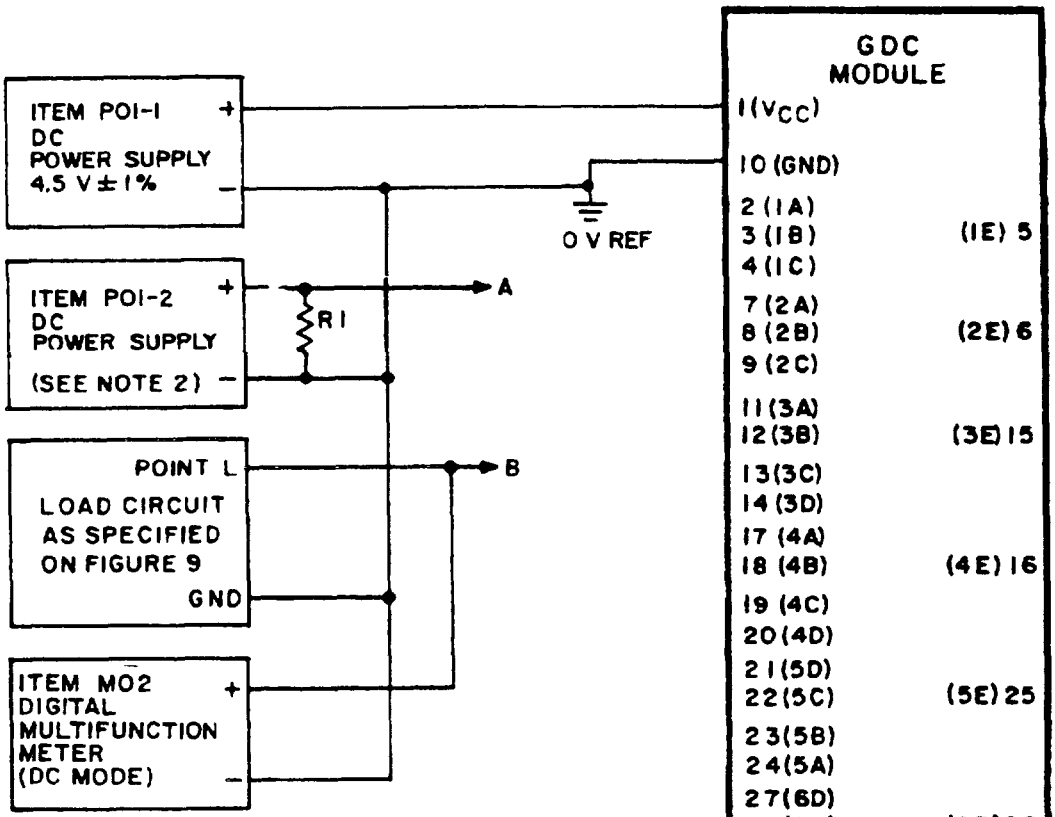
Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
D1	1E	Output voltage, low level	2,3,4	5	0	450	mV	8
D2	2E		7,8,9	6				
D3	3E		11,12,13,14	15				
D4	4E		17,18,19,20	16				
D5	5E		21,22,23,24	25				
D6	6E		27,28,29,30	26				
D7	7E		31,32,33,34	35				
D8	8E		37,38,39,40	36				
E1	1E	Output voltage, high level	2	5	2.4	4.5	V	
E2	1E		3	5				
E3	1E		4	5				
E4	2E		7	6				
E5	2E		8	6				
E6	2E		9	6				
E7	3E		11	15				
E8	3E		12					
E9	3E		13					
E10	3E		14					
E11	4E		17	16				
E12	4E		18	16				

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* TABLE VI. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E13	4E Output voltage, high level	19	16	2.4	4.5	V	8
E14	4E	20	16				
E15	5E	21	25				
E16	5E	22					
E17	5E	23					
E18	5E	24					
E19	6E	27	26				
E20	6E	28					
E21	6E	29					
E22	6E	30					
E23	7E	31	35				
E24	7E	32					
E25	7E	33					
E26	7E	34					
E27	8E	37	36				
E28	8E	38					
E29	8E	39					
E30	8E	40					

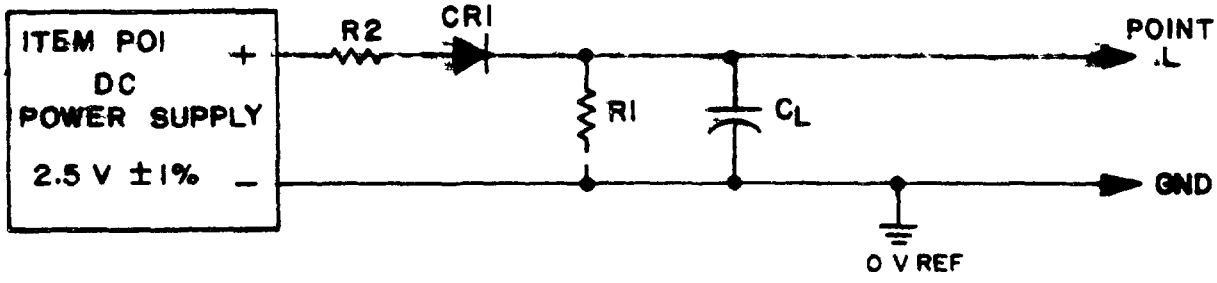
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NOTES
1. R1 = 49.9Ω ±1%, 1 W.
2. Set item POI-2 voltage as follows:
   V_{IN} = 2.0 V ±1% for test conditions D1-D8.
   V_{IN} = 0.8 V ±1% for test conditions E1-E30.

* FIGURE 8. Output voltage (low-level and high-level) test circuit for table VI.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. R1 = 806Ω ±1%, 1/8 W.
3. R2 = Select value for a current flow of 44 mA ±1% into point L with point L to ground.
4. C_L = 1,000 ±5 pF (including probe and parasitic capacitance) for test conditions K, L, M, and N.
5. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions R and S.
6. Capacitor C_L is not required for dc electrical test conditions D and E.
7. During output switching the 2.5 V power supply shall have a maximum overshoot of 500 mV peak.

FIGURE 9. GDC logic load circuit.

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* TABLE VII. Input current (low-level and high-level) and output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1A Input current, low level	3,4	2	Open	1.0	2.3	mA	10
F2	1B	2,4	3					
F3	1C	2,3	4					
F4	2A	8,9	7					
F5	2B	7,9	8					
F6	2C	7,8	9					
F7	3A	12,13,14	11					
F8	3B	11,13,14	12					
F9	3C	11,12,14	13					
F10	3D	11,12,13	14					
F11	4A	18,19,20	17					
F12	4B	17,19,20	18					
F13	4C	17,18,20	19					
F14	4D	17,18,19	20					
F15	5D	22,23,24	21					
F16	5C	21,23,24	22					
F17	5B	21,22,24	23					
F18	5A	21,22,23	24					
F19	6D	28,29,30	27					
F20	6C	27,29,30	28					
F21	6B	27,28,30	29					
F22	6A	27,28,29	30					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:24Z
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* TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F23	7D	Input current, low level	32,33,34	31	Open	1.0	2.3	mA	10
F24	7C		31,33,34	32					
F25	7B		31,32,34	33					
F26	7A		31,32,33	34					
F27	8D		38,39,40	37					
F28	8C		37,39,40	38					
F29	8B		37,38,40	39					
F30	8A		37,38,39	40					
G1	1A	Input current, high level	2	Open	3	0	120	µA	
G2	1A		2		4				
G3	1B		3		2				
G4	1B		3		4				
G5	1C		4		2				
G6	1C		4		3				
G7	2A		7		8				
G8	2A		7		9				
G9	2B		8		7				
G10	2B		8		9				
G11	2C		9		7				
G12	2C		9		8				
G13	3A		11		12				

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* TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G13	3A Input current, high level	11	Open	12	0	120	µA	10
G14	3A	11		13				
G15	3A	11		14				
G16	3B	12		11				
G17	3B	12		13				
G18	3B	12		14				
G19	3C	13		11				
G20	3C	13		12				
G21	3C	13		14				
G22	3D	14		11				
G23	3D	14		12				
G24	3D	14		13				
G25	4A	17		18				
G26	4A	17		19				
G27	4A	17		20				
G28	4B	18		17				
G29	4B	18		19				
G30	4B	18		20				
G31	4C	19		17				
G32	4C	19		18				
G33	4C	19		20				

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* TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G34	4D Input current, high level	20	Open	17	0	120	µA	10
G35	4D	20		18				
G36	4D	20		19				
G37	5D	21		22				
G38	5D	21		23				
G39	5D	21		24				
G40	5C	22		21				
G41	5C	22		23				
G42	5C	22		24				
G43	5B	23		21				
G44	5B	23		22				
G45	5B	23		24				
G46	5A	24		21				
G47	5A	24		22				
G48	5A	24		23				
G49	6D	27		28				
G50	6D	27		29				
G51	6D	27		30				
G52	6C	28		27				
G53	6C	28		29				
G54	6C	28		30				
G55	6B	29		27				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

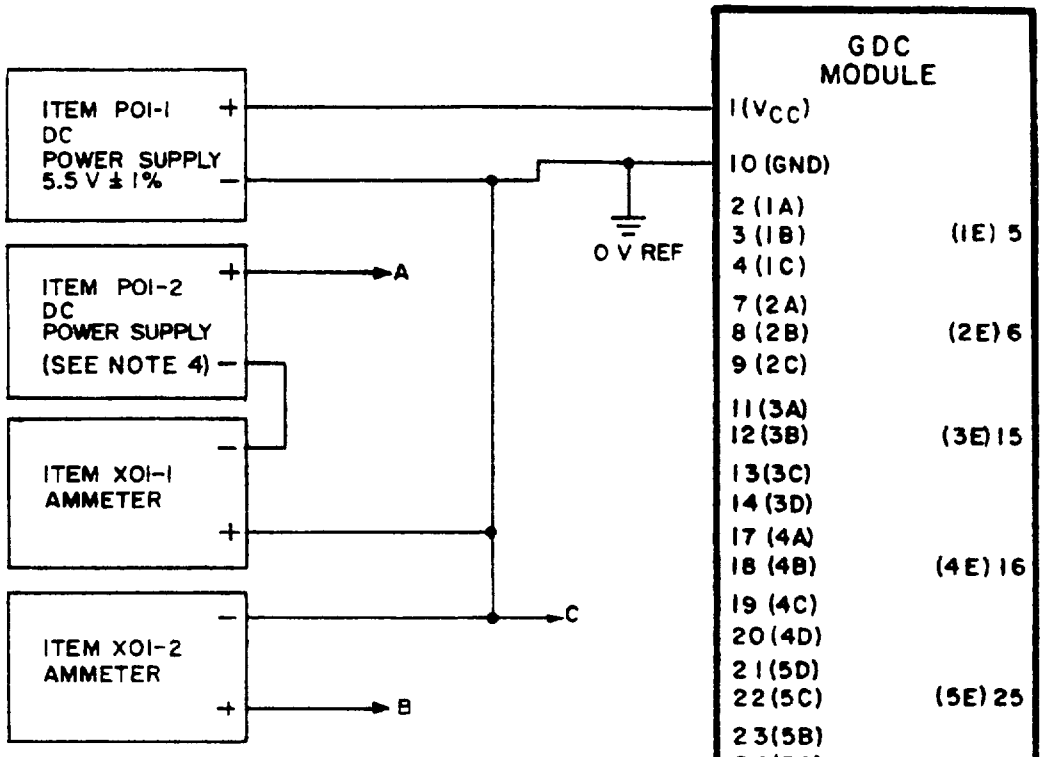
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G56	6B Input current, high level	29	Open	28	0	120	µA	10
G57	6B	29		30				
G58	6A	30		27				
G59	6A	30		28				
G60	6A	30		29				
G61	7D	31		32				
G62	7D	31		33				
G63	7D	31		34				
G64	7C	32		31				
G65	7C	32		33				
G66	7C	32		34				
G67	7B	33		31				
G68	7B	33		32				
G69	7B	33		34				
G70	7A	34		31				
G71	7A	34		32				
G72	7A	34		33				
G73	8D	37		38				
G74	8D	37		39				
G75	8D	37		40				
G76	8C	38		37				
G77	8C	38		39				

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* TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G78	8C	Input current, high level	38	Open	40	0	120	µA	10
G79	8B		39		37				
G80	8B		39		38				
G81	8B		39		40				
G82	8A		40		37				
G83	8A		40		38				
G84	8A		40		39				
H1	1E	Output short circuit current	Open	5	2,3,4	50	165	mA	
H2	2E			6	7,8,9				
H3	3E			15	11,12,13,14				
H4	4E			16	17,18,19,20				
H5	5E			25	21,22,23,24				
H6	6E			26	27,28,29,30				
H7	7E			35	31,32,33,34				
H8	8E			36	37,38,39,40				

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTES:
1. Since items XO1-1 and XO1-2 are not used simultaneously, only one item XO1 is required.
2. With point A disconnected, item XO1-1 should indicate 0.0 current on the 1.0 µA range.
3. When performing output short circuit current test, only one output shall be shorted at a time, and the measurement taken within 5 seconds.
4. Set item PO1-2 voltage as follows:
   V_{IN} = 4.5 V ±1% for test conditions F1-F30.
   V_{IN} = 2.4 V ±1% for test conditions G1-G84.

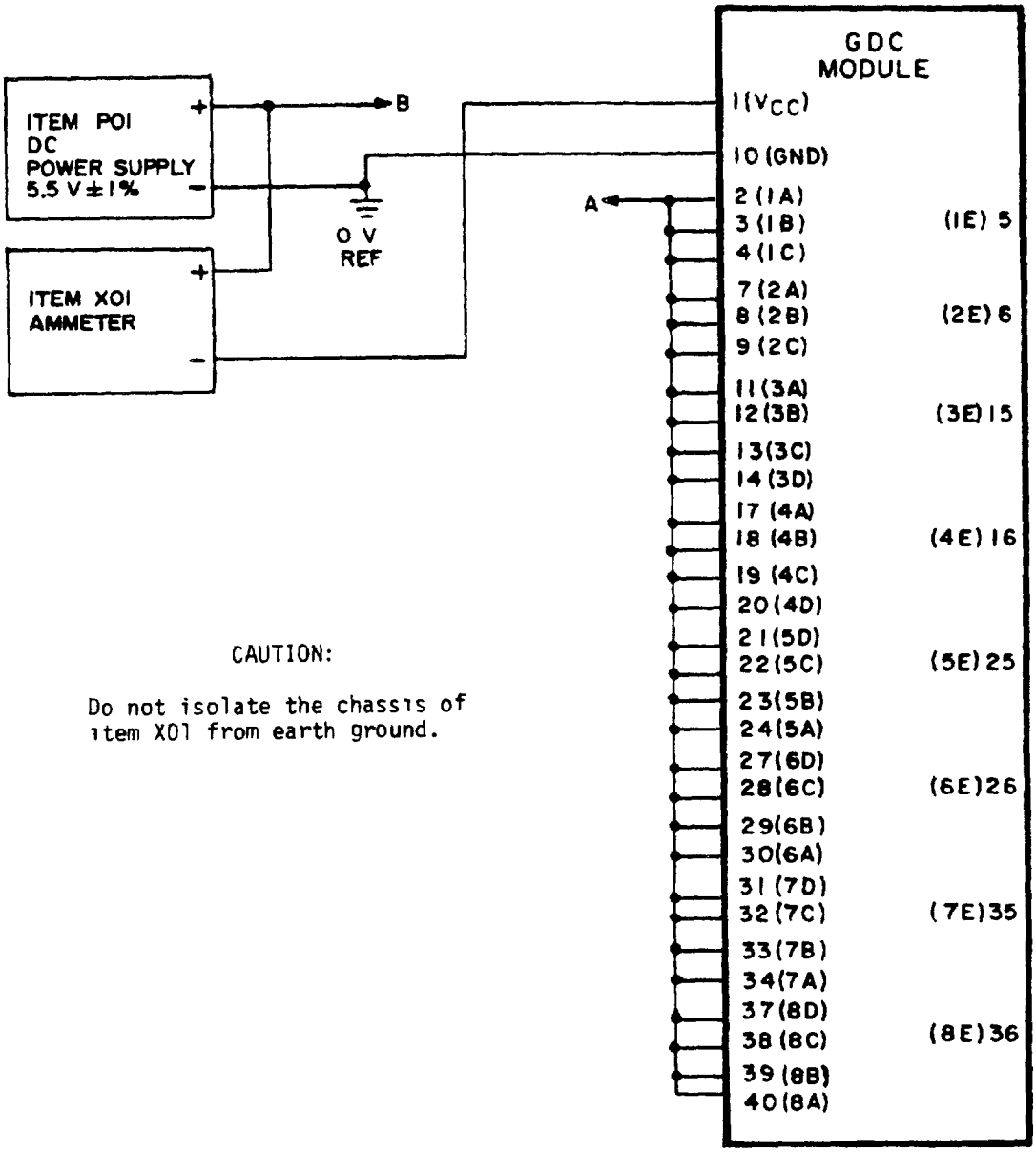
* FIGURE 10. Input current (low-level and high-level) and output short circuit current test circuit for table VII.

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TABLE VIII. Power supply current (low-level and high level) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
J1	Power supply current, low level	B	60	124	mA	11
J2	Power supply current, high level	Pin 10	10	40	mA	11

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

* FIGURE 11. Power supply current (low-level and high-level) test circuit for table VIII.

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* TABLE IX. Transition time (t_{THL}) test.

Test condition	Parameter	Connect 1/		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	1E Transition time (t_{THL})	2	5	3	45	ns	12,13
K2	2E	7	6				
K3	3E	11	15				
K4	4E	17	16				
K5	5E	21	25				
K6	6E	27	26				
K7	7E	31	35				
K8	8E	37	36				

1/ Connect pins 2, 7, 11, 17, 21, 27, 31, and 37 to pin 10, excluding from this group the pin connected to point A for each test condition.

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* TABLE X. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect 1/		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	1E Transition time (t_{TLH})	2	5	3	45	ns	12,13
L2	2E	7	6				
L3	3E	11	15				
L4	4E	17	16				
L5	5E	21	25				
L6	6E	27	26				
L7	7E	31	35				
L8	8E	37	36				

1/ Connect pins 2, 7, 11, 17, 21, 27, 31, and 37 to pin 10, excluding from the pin this group the pin connected to point A for each test condition.

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* TABLE XI. Propagation delay time (t_{PLH}) test.

Test condition	Parameter		Connect 1/		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
M1	1E	Propagation delay time (t_{PLH})	2	5	3	40	ns	12,13
M2	2E		7	6				
M3	3E		11	15				
M4	4E		17	16				
M5	5E		21	25				
M6	6E		27	26				
M7	7E		31	35				
M8	8E		37	36				

1/ Connect pins 2, 7, 11, 17, 21, 27, 31, and 37 to pin 10, excluding from this group the pin connected to point A for each test condition.

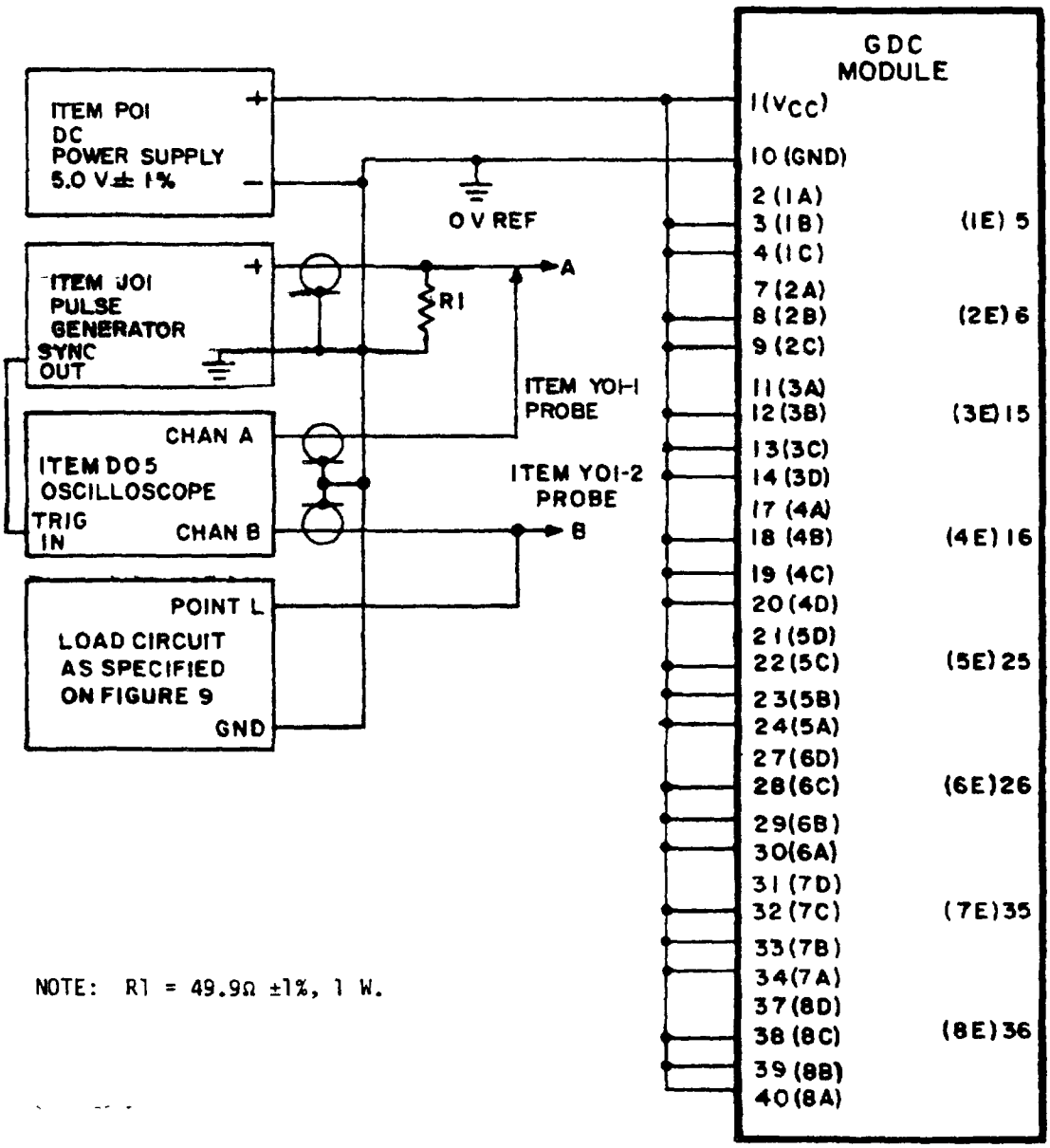
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TABLE XII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect 1/		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	1E Propagation delay time (t_{PHL})	2	5	3	50	ns	12,13
N2	2E	7	6				
N3	3E	11	15				
N4	4E	17	16				
N5	5E	21	25				
N6	6E	27	26				
N7	7E	31	35				
N8	8E	37	36				

1/ Connect pins 2, 7, 11, 17, 21, 27, 31, and 37 to pin 10, excluding from this group the pin connected to point A for each test condition.

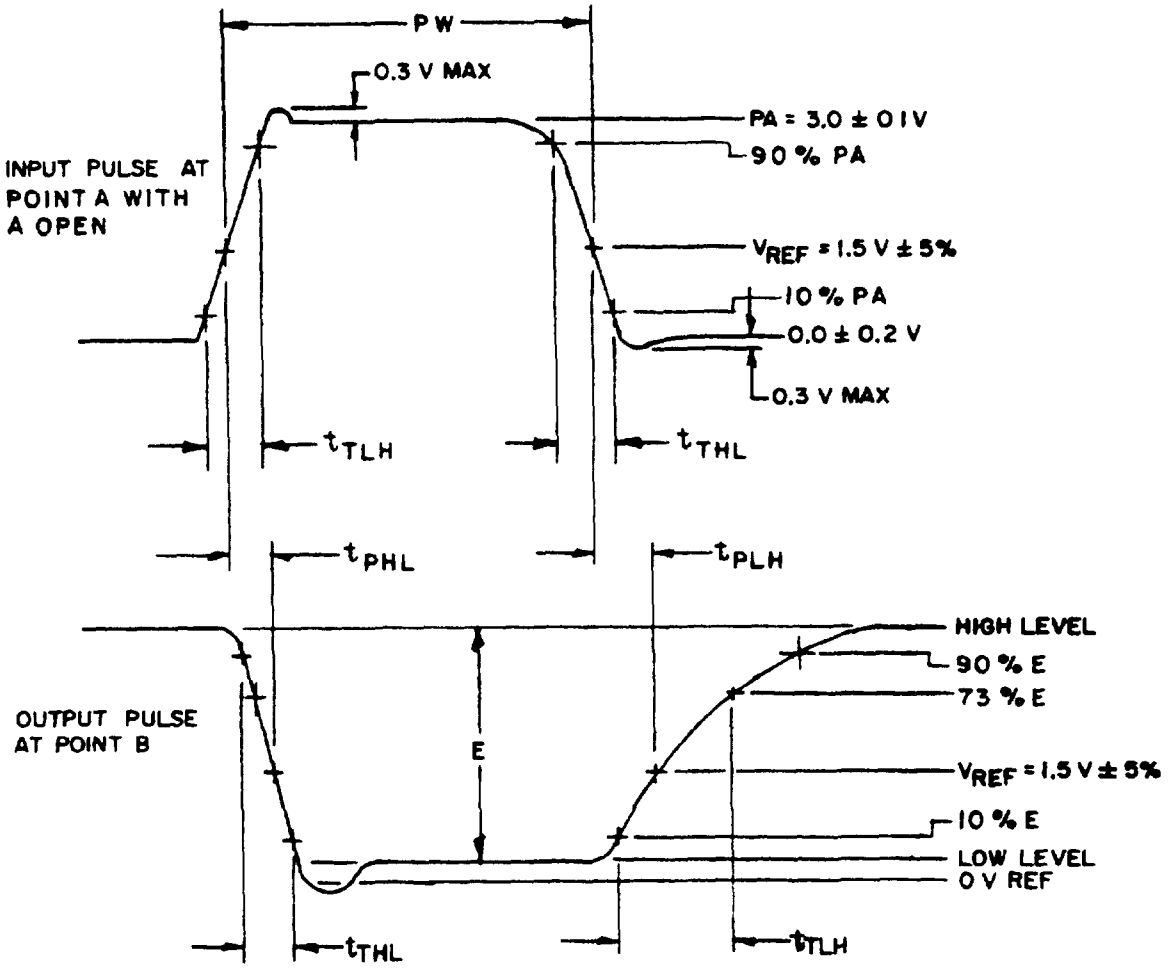
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NOTE: R1 = 49.9Ω ±1%, 1 W.

* FIGURE 12. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables IX through XII.

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NOTES:
1. Input pulse parameters.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PW = 500 ±50 ns.
   PRR = 1.0 MHz ±10%.
2. All voltages are referenced to ground.
3. Before making measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.45 V.

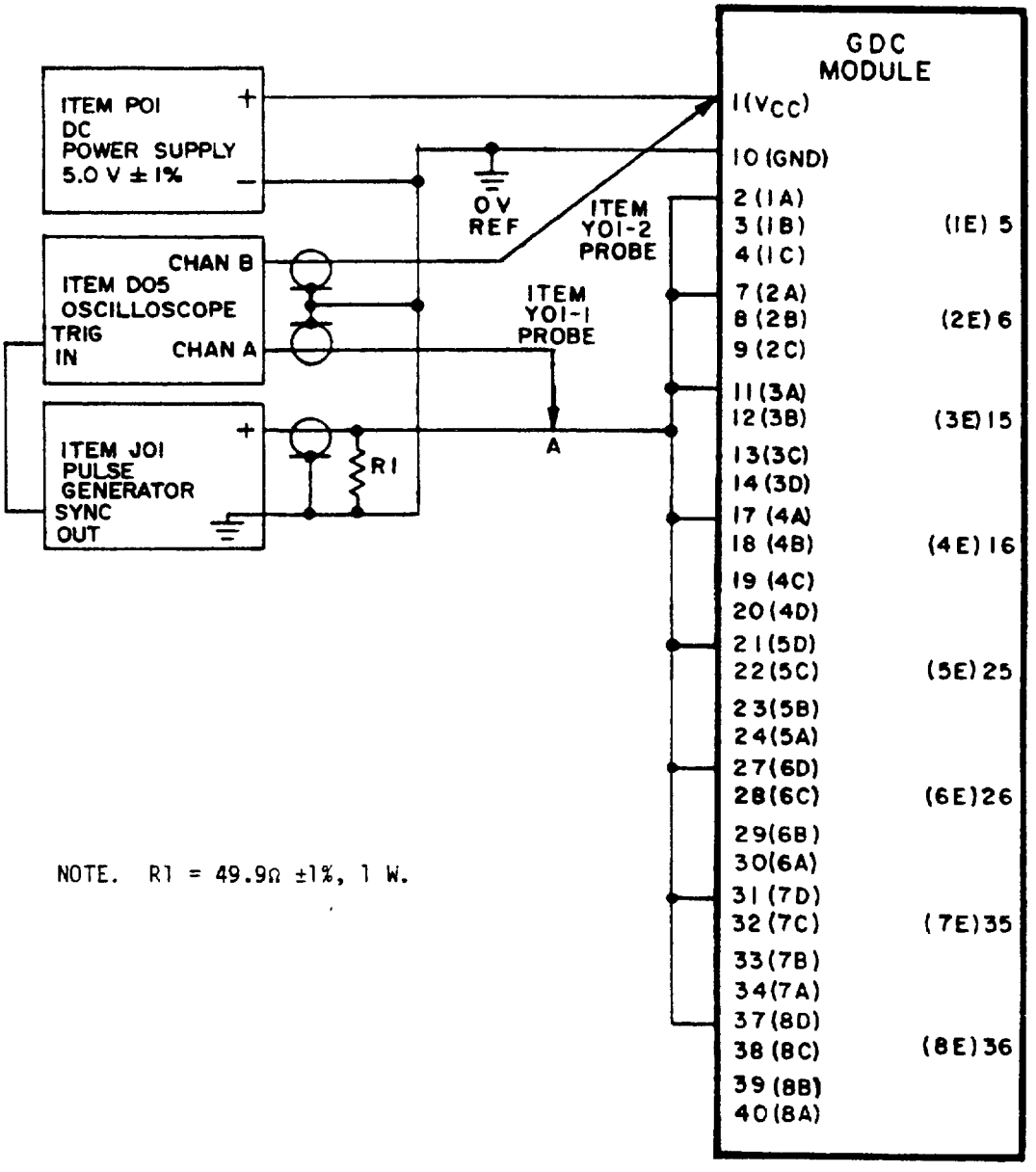
* FIGURE 13. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables IX through XII.

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TABLE XIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
P	Power supply transient amplitude	0	500	mV	14.15

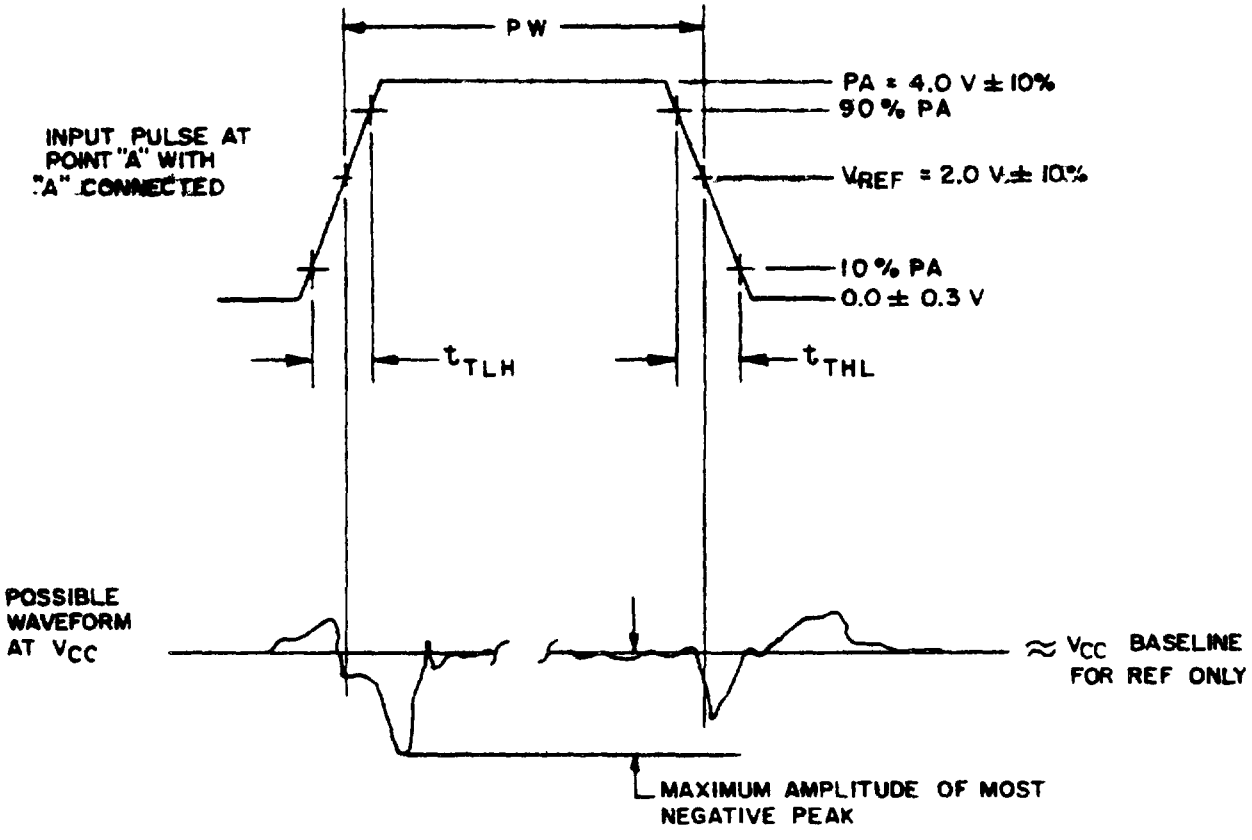
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NOTE. R1 = 49.9Ω ±1%, 1 W.

* FIGURE 14. Power supply transient amplitude test circuit for table XIII.

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NOTES:
1. The most negative spike must comply with test condition P of table XIII.
2. Input pulse parameters:
   PW = 3 µs ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±1 ns.
   PRR = 100 kHz ±10%.

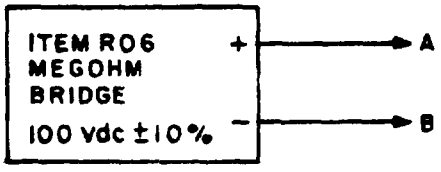
* FIGURE 15. Power supply transient amplitude test circuit waveforms for table XIII.

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* TABLE XIV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q	Isolation	All pins	Frame	10	∞	MΩ	16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:24Z
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GDC MODULE	
1 (V_{CC})	
10 (GND)	
2 (1A)	
3 (1B)	(1E) 5
4 (1C)	
7 (2A)	
8 (2B)	(2E) 6
9 (2C)	
11 (3A)	
12 (3B)	(3E) 15
13 (3C)	
14 (3D)	
17 (4A)	
18 (4B)	(4E) 16
19 (4C)	
20 (4D)	
21 (5D)	
22 (5C)	(5E) 25
23 (5B)	
24 (5A)	
27 (6D)	
28 (6C)	(6E) 26
29 (6B)	
30 (6A)	
31 (7D)	
32 (7C)	(7E) 35
33 (7B)	
34 (7A)	
37 (8D)	
38 (8C)	(8E) 36
39 (8B)	
40 (8A)	

WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 16. Isolation test circuit for table XIV.

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TABLE XV. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	Output crosstalk, low level	E	F,G,H,I,J,K,L	5	Open	0	800	mV	17,18
R2		F	E,G,H,I,J,K,L	6					
R3		G	E,F,H,I,J,K,L	15					
R4		H	E,F,G,I,J,K,L	16					
R5		I	E,F,G,H,J,K,L	25					
R6		J	E,F,G,H,I,K,L	26					
R7		K	E,F,G,H,I,J,L	35					
R8		L	E,F,G,H,I,J,K	36					
R9		E,F,I,J	G,H,K,L	5			400		
R10				6					
R11				25					
R12				26					
R13		E,G,I,K	F,H,J,L	5					
R14				15					
R15				25					
R16				35					
R17		E,H,I,L	F,G,J,K	5					
R18				16					
R19				25					
R20				36					

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* TABLE XV. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R21	Output crosstalk, low level	F,H,J,L	E,G,I,K	6	Open	0	400	mV	17,18
R22				16					
R23				26					
R24				36					
R25		F,G,J,K	E,H,I,L	6					
R26				15					
R27				26					
R28				35					
R29		G,H,K,L	E,F,I,J	15					
R30				16					
R31				35					
R32				36					
S1	Output crosstalk, high level	Open	F,G,H,I,J,K,L	5	E		800		
S2			E,G,H,I,J,K,L,	6	F				
S3			E,F,H,I,J,K,L	15	G				
S4			E,F,G,I,J,K,L	16	H				
S5			E,F,G,H,J,K,L	25	I				
S6			E,F,G,H,I,K,L	26	J				
S7			E,F,G,H,I,J,L	35	K				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:24Z
Check the source to verify that this is the current version before use.

* TABLE XV. Output crosstalk (low-level and high-level) test - Continued.

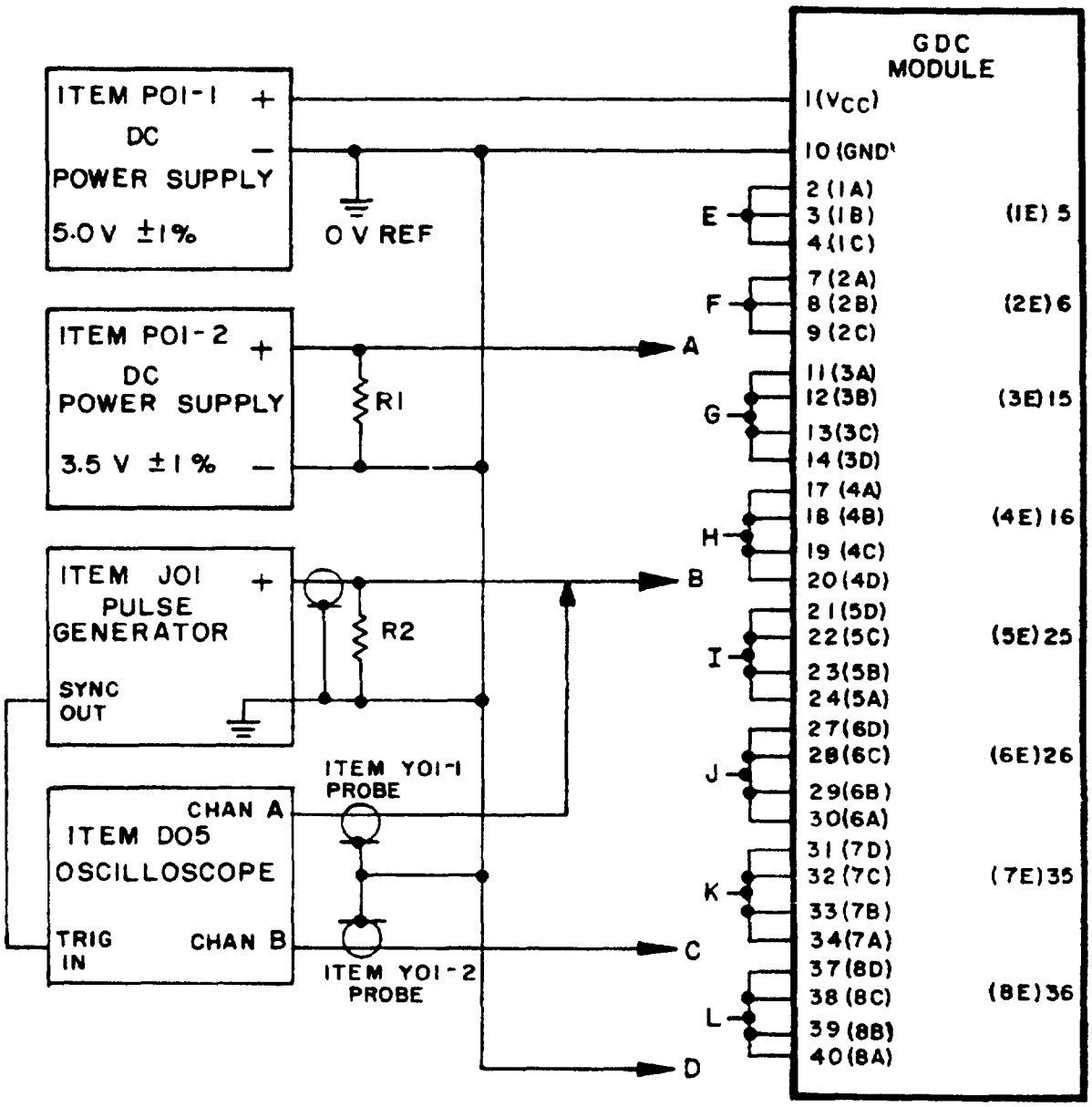
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S8	Output crosstalk, high level	Open	E,F,G,H,I,J,K	36	L	0	800	mV	17,18
S9			G,H,K,L	5	E,F,I,J		400		
S10				6					
S11				25					
S12				26					
S13			F,H,J,L	5	E,G,I,K				
S14				15					
S15				25					
S16				35					
S17			F,G,J,K	5	E,H,I,L				
S18				16					
S19				25					
S20				36					
S21			E,G,I,K	6	F,H,J,L				
S22				16					
S23				26					
S24				36					
S25			E,H,I,L	6	F,G,J,K				
S26				15					
S27				26					
S28				35					
S29			L,F,I,J	15	G,H,K,L				

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* TABLE XV. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S30	Output crosstalk, high level	Open	E,F,I,J,	16	G,H,K,L	0	400	mV	17,18
S31	Output crosstalk, high level	Open	E,F,I,J	35	G,H,K,L	0	400	mV	17,18
S32	Output crosstalk, high level	Open	E,F,I,J	36	G,H,K,L	0	400	mV	17,18

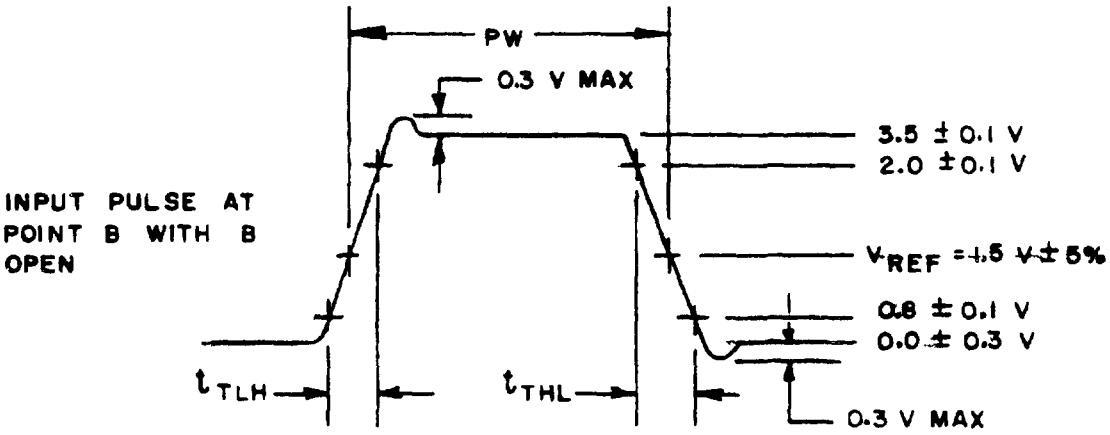
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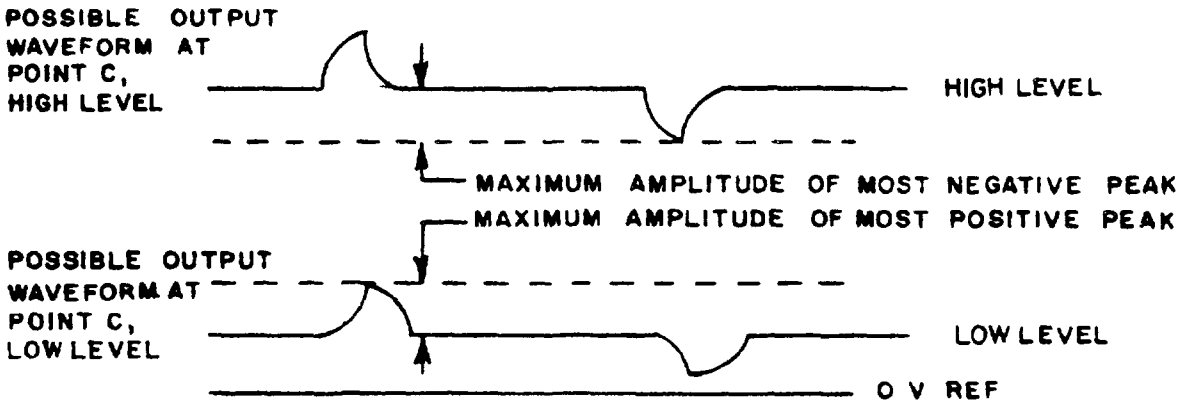


NOTES
1. R1 = R2 = 49.9Ω ±1%, 1 W.
2. Load all outputs with load as specified on figure 9.

* FIGURE 17. Output crosstalk (low-level and high-level) test circuit for table XV.

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NOTE: Input pulse parameters:
PW = 200 ns ±10%.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
PRR = 1.0 MHz ±10%.

FIGURE 10. Output crosstalk (low level and high level) test.

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* TABLE XVI. Input capacitance test.

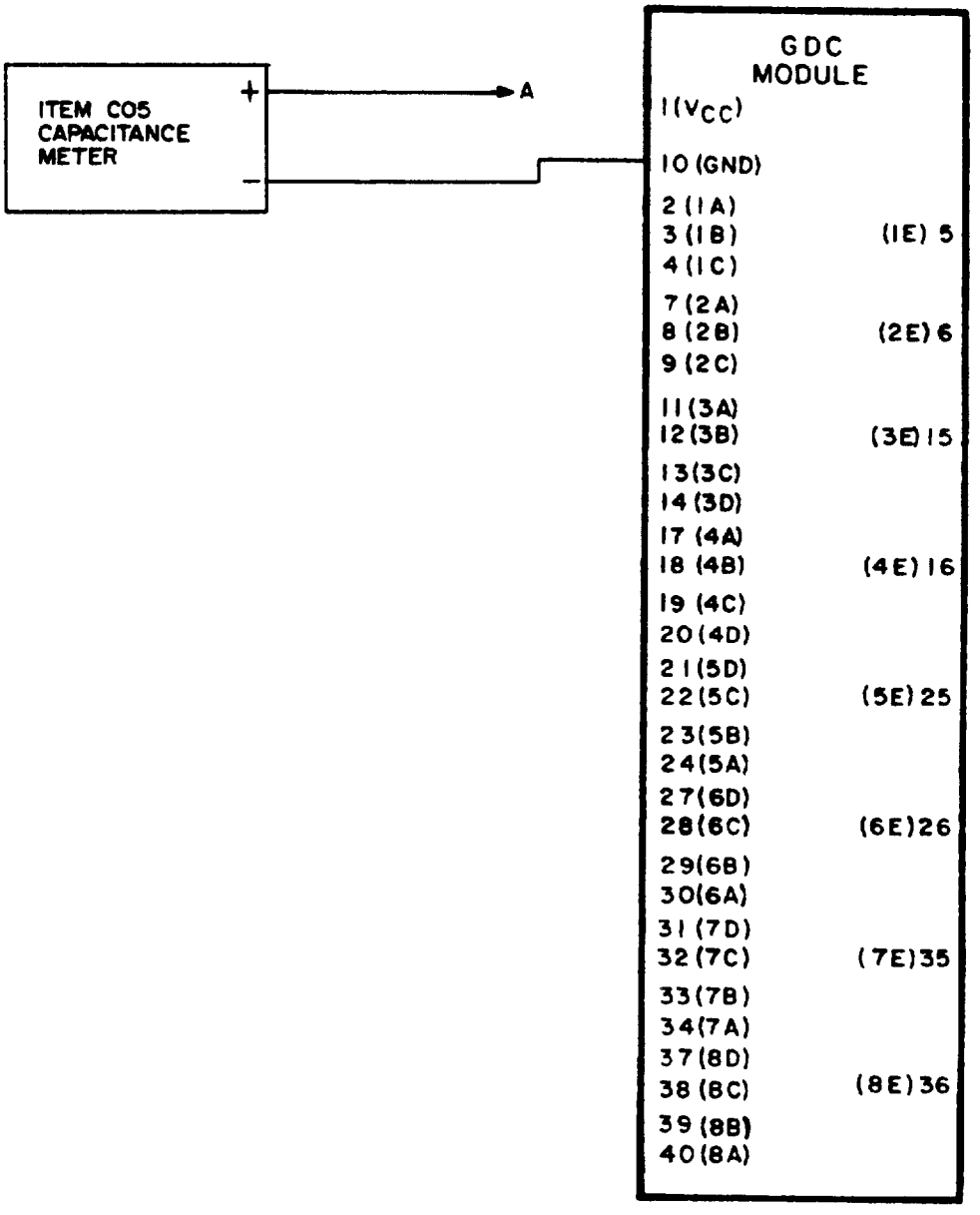
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
T1	1A Input capacitance	2	2	15	pF	19
T2	1B	3				
T3	1C	4				
T4	2A	7				
T5	2B	8				
T6	2C	9				
T7	3A	11				
T8	3B	12				
T9	3C	13				
T10	3D	14				
T11	4A	17				
T12	4B	18				
T13	4C	19				
T14	4D	20				
T15	5D	21				
T16	5C	22				
T17	5B	23				

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* TABLE XVI. Input capacitance test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
T18	5A Input capacitance	24	2	15	pF	19
T19	6D	27				
T20	6C	28				
T21	6B	29				
T22	6A	30				
T23	7D	31				
T24	7C	32				
T25	7B	33				
T26	7A	34				
T27	8D	37				
T28	8C	38				
T29	8B	39				
T30	8A	40				

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NOTES.
1 Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

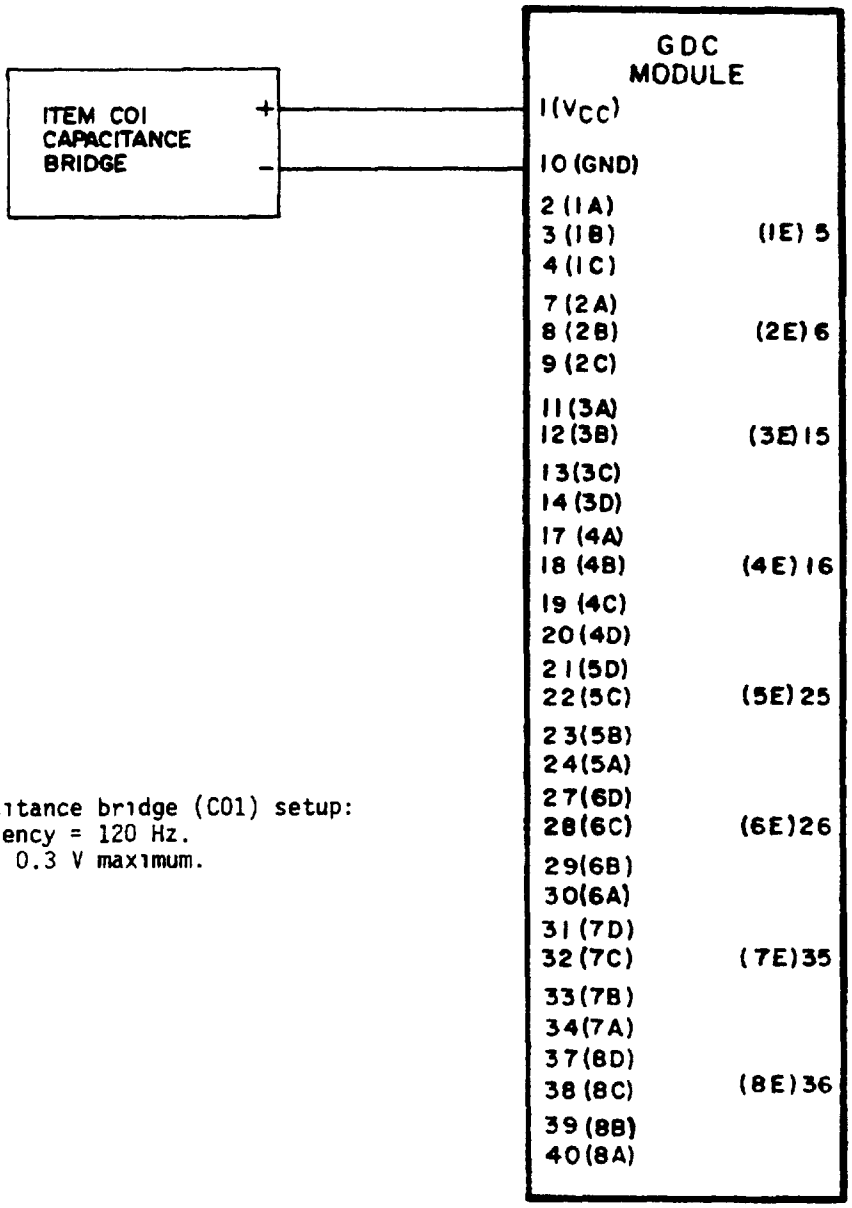
* FIGURE 19. Input capacitance test circuit for table XVI.

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TABLE XVII. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply filter capacitance	0.007	3.0	µF	20

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NOTE: Capacitance bridge (C01) setup:
Frequency = 120 Hz.
V_{pk} = 0.3 V maximum.

* Figure 20. Power supply filter capacitance test circuit for table XVII.

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